

**Concordia University
Faculty of Fine Arts**

**Generative AI
CART-498**

**Assignment 2
Reflection**

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Reflection

Altering the X value was an interesting experiment as every value always produces a unique output. I began my experimentation from $P+1$, finding the poem in which the words were replaced by the first highest prediction, and increased the value. It was visible that the output of $P+X$ poems with higher X values would find increasingly unexpected or seemingly random replacement words. After experimenting with different values, I tried increasing the X values to the thousands and tens of thousands. I looked up how many tokens were in GPT-2's vocabulary, I then proceeded to output a $P+50,257$ poem to see what was the poem with the least probable words as replacements. Many lines would contain non-printable characters, so I added some code to exclude these characters and default to the next least probable printable word. As a result I got this poem with the least probable words, which was the one I chose as my second version.

To build an algorithm that would replace every noun in the poem, I would find a library or function to help me identify nouns. For each line of the poem, the algorithm would scan each word, when finding a noun, it would use the words that preceded it in that line as context, and would then use it to find the X -highest probability prediction. This would be interesting as lines with multiple nouns would see their first nouns have an impact on the prediction of the subsequent noun replacements.